0.1 Solution

We are given $k \geq 3$ distinct positive integers $a_1, \dots, a_k$ with product P and $\sum a_i < 3 \cdot 10^6$. Define g as the greatest common divisor of the numbers $\frac{P}{a_i} + 1$ $(1 \leq i \leq k)$. We want the maximum possible value of g over all choices of $k \geq 3$ and a_i.

0.1.1 1. Basic properties of g

Let $N_i = \frac{P}{a_i} + 1$. Take two indices $i \neq j$. Because $g \mid N_i$ and $g \mid N_j$, we have

$$g \mid N_i - N_j = \Big(\prod_{t \neq i} a_t\Big) - \Big(\prod_{t \neq j} a_t\Big) = (a_j - a_i) \prod_{t \neq i,j} a_t.$$

We first show that $\gcd(g, a_t) = 1$ for every t. Suppose a prime p divides both g and some a_t. Consider the number $N_t = \prod_{u \neq t} a_u + 1$. Since $p \mid a_t$ and $p \mid g$, the congruence $N_t \equiv 0 \pmod p$ gives $\prod_{u \neq t} a_u \equiv -1 \pmod p$. For any $i \neq t$, $N_i = a_t \cdot \prod_{u \neq i,t} a_u + 1 \equiv 1 \pmod p$ (because $a_t \equiv 0 \pmod p$). But $p \mid g$ and $g \mid N_i$, so $p \mid N_i$, contradicting $N_i \equiv 1 \pmod p$. Hence $\gcd(g, a_t) = 1$ for all t.

Consequently $\gcd\big(g, \prod_{t \neq i,j} a_t\big) = 1$, and from $g \mid (a_j - a_i) \prod_{t \neq i,j} a_t$ we deduce

$$g \mid a_j - a_i \qquad \text{for all } i, j.$$

Thus all a_i are congruent modulo g. Write

$$a_i = r + m_i g \qquad (i = 1, \dots, k)$$

with $1 \leq r \leq g - 1$ (since $\gcd(r, g) = 1$ follows from $\gcd(g, a_i) = 1$) and the m_i are distinct non-negative integers.

Now reduce N_i modulo g. Because $a_j \equiv r \pmod g$ for all j, we have

$$\prod_{t \neq i} a_t \equiv r^{k-1} \pmod g.$$

Since $g \mid N_i = \prod_{t \neq i} a_t + 1$, we obtain

$$r^{k-1} + 1 \equiv 0 \pmod g \quad \Longrightarrow \quad g \mid r^{k-1} + 1.$$

0.1.2 2. Why $k = 3$ gives the largest possible g

For fixed r, g the sum $S = \sum a_i$ is at least the sum with the smallest possible distinct non-negative m_i, namely $0, 1, \dots, k-1$:

$$S \geq kr + g \cdot \frac{k(k-1)}{2}.$$

If we try to make g large we want S as small as possible, so we take this minimal sum.

For $k = 3$ this gives $S_{\min} = 3r + 3g$. For $k = 4$ we get $S_{\min} = 4r + 6g$, and for larger k the g-term grows quadratically. Since $S < 3 \cdot 10^6$, we obtain for $k = 4$: $6g < 3 \cdot 10^6 \Rightarrow g < 500\,000$. For $k = 5$ the bound becomes even smaller. Thus the maximum possible g can only be achieved when $k = 3$ (the smallest allowed k).

From now on we set $k = 3$. Then condition (1) becomes

$$g \mid r^2 + 1.$$

The minimal sum condition $3(r + g) < 3 \cdot 10^6$ is equivalent to

$$r + g < 10^6.$$

Conversely, if (2) and (3) hold we can simply take

$$a_1 = r, \quad a_2 = r + g, \quad a_3 = r + 2g,$$

which are distinct, positive, and satisfy the sum bound. Moreover, modulo g we have $a_i \equiv r$, so

$$\frac{P}{a_i} + 1 \equiv r^2 + 1 \equiv 0 \pmod{g},$$

hence g divides each $\frac{P}{a_i} + 1$. As argued in §1, any common divisor must divide the differences $a_j - a_i$, which are g and $2g$, so the gcd of the three numbers is exactly g.

Therefore the problem reduces to:

> Find the largest integer g for which there exist positive integers r, g with
>
> $$1 \le r < g, \qquad g \mid r^2 + 1, \qquad r + g < 10^6.$$

0.1.3 3. Maximising g

Write $r^2 + 1 = g\,h$ with $h \in \mathbb{N}$. Then $g = \dfrac{r^2 + 1}{h}$ and because $g > r$ we have $h \le r$. Condition (3) becomes

$$r + \frac{r^2 + 1}{h} < 10^6 \iff r^2 + hr + 1 < h \cdot 10^6.$$

For a given h, (4) gives an upper bound on r. To make g as large as possible we want h as small as possible (since g decreases when h increases). We examine the smallest admissible values of h.

- $h = 1$**.** Then $g = r^2 + 1$ and (4) reads $r^2 + r + 1 < 10^6$. The largest integer r satisfying this is $r = 999$ (for $r = 1000$ we get $1\,001\,001 > 10^6$). This yields $g = 999^2 + 1 = 998\,001$.

- $h = 2$**.** Here r must be odd so that $r^2 + 1$ is even. Then $g = \frac{r^2+1}{2}$ and (4) becomes

  $$r + \frac{r^2 + 1}{2} < 10^6 \iff (r + 1)^2 < 2 \cdot 10^6.$$

  The largest integer r with $(r + 1)^2 < 2\,000\,000$ is $r = 1413$ (since $1414^2 = 1\,999\,396$ but $1415^2 = 2\,002\,025$). Because r must be odd, $r = 1413$ is admissible. Then

  $$g = \frac{1413^2 + 1}{2} = \frac{1\,996\,569 + 1}{2} = \frac{1\,996\,570}{2} = 998\,285.$$

  This is larger than $998\,001$.

- $h = 3$**.** $r^2 + 1$ is never divisible by 3 because a square modulo 3 is 0 or 1, so $r^2 + 1 \equiv 1$ or $2 \pmod 3$. Hence $h = 3$ is impossible.

- $h = 4$**.** $r^2 + 1 \equiv 0 \pmod 4$ would require $r^2 \equiv 3 \pmod 4$, which is impossible (squares are 0 or 1 modulo 4). So $h = 4$ impossible.

- $h = 5$. This is possible when $r \equiv \pm 2 \pmod 5$. Condition (4) gives $r^2 + 5r + 1 < 5 \cdot 10^6$, i.e. $r^2 + 5r - 4\,999\,999 < 0$, whose positive root is approximately 2233.5, so $r \le 2233$. The resulting g is at most $\frac{2233^2+1}{5} \approx 997\,258$, which is smaller than $998\,285$.
- For any $h \ge 5$ the bound on r becomes even tighter and the maximum possible g is even smaller.

Consequently the overall maximum g is obtained with $h = 2$, $r = 1413$, giving $g = 998\,285$.

0.1.4 4. Construction achieving $g = 998\,285$

Take $k = 3$, $r = 1413$, $g = 998\,285$. Define

$$a_1 = 1413, \qquad a_2 = 1413 + 998\,285 = 999\,698, \qquad a_3 = 1413 + 2 \cdot 998\,285 = 1\,997\,983.$$

These are distinct positive integers. Their sum is

$$1413 + 999\,698 + 1\,997\,983 = 2\,999\,094 < 3 \cdot 10^6.$$

Their product $P = a_1 a_2 a_3$ is huge, but we only need to check that g divides each $\frac{P}{a_i} + 1$. Modulo g we have $a_i \equiv r = 1413$, so

$$\frac{P}{a_i} = \prod_{j \ne i} a_j \equiv r^2 = 1413^2 \pmod g.$$

Hence

$$\frac{P}{a_i} + 1 \equiv r^2 + 1 \equiv 0 \pmod g,$$

because $r^2 + 1 = 1\,996\,570 = 2 \cdot 998\,285$ is a multiple of g. As argued in §1, any common divisor of the three numbers must divide the differences $a_j - a_i$, which are g and $2g$, and since g itself divides all of them, the gcd is exactly g.

Thus $g = 998\,285$ is attainable.

0.1.5 5. Upper bound

We have shown that any feasible configuration with $k = 3$ must satisfy $g \mid r^2 + 1$ and $r + g < 10^6$. The analysis in §3 shows that for any such r, g, the value g cannot exceed $998\,285$ (the maximum obtained for $h = 2$, $r = 1413$). For $k \ge 4$ we already proved $g < 500\,000$, which is smaller. Hence $998\,285$ is the absolute maximum possible value of g.

Answer: $\boxed{998285}$